

Bioequivalence of Docosahexaenoic acid from different algal oils in capsules and in a DHA-fortified food.

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Docosahexaenoic acid (DHA), a long-chain omega-3 fatty acid, is important for eye and brain development and ongoing visual, cognitive, and cardiovascular health. Unlike fish-sourced oils, the bioavailability of DHA from vegetarian-sourced (algal) oils has not been formally assessed. We assessed bioequivalence of DHA oils in capsules from two different algal strains versus bioavailability from an algal-DHA-fortified food. Our 28-day randomized, placebo-controlled, parallel group study compared bioavailability of (a) two different algal DHA oils in capsules ("DHASCO-T" and "DHASCO-S") at doses of 200, 600, and 1,000 mg DHA per day (n = 12 per group) and of (b) an algal-DHA-fortified food (n = 12). Bioequivalence was based on changes in plasma phospholipid and erythrocyte DHA levels. Effects on arachidonic acid (ARA), docosapentaenoic acid-n-6 (DPAn-6), and eicosapentaenoic acid (EPA) were also determined. Both DHASCO-T and DHASCO-S capsules produced equivalent DHA levels in plasma phospholipids and erythrocytes. DHA response was dose-dependent and linear over the dose range, plasma phospholipid DHA increased by 1.17, 2.28 and 3.03 g per 100 g fatty acid at 200, 600, and 1,000 mg dose, respectively. Snack bars fortified with DHASCO-S oil also delivered equivalent amounts of DHA on a DHA dose basis. Adverse event monitoring revealed an excellent safety and tolerability profile. Two different algal oil capsule supplements and an algal oil-fortified food represent bioequivalent and safe sources of DHA.